

Troi: Towards Understanding Users Perspectives to Mobile Automatic Emotion Recognition System in Their Natural Setting

VIPULA DISSANAYAKE, Augmented Human Lab, Auckland Bioengineering Institute (ABI), The University of Auckland, New Zealand

VANESSA TANG, Augmented Human Lab, ABI, The University of Auckland, New Zealand

DON SAMITHA ELVITIGALA, HCI BoDi Lab, School of Computer Science and Engineering, University of New South Wales, Australia

ELLIOTT WEN, Augmented Human Lab, ABI, The University of Auckland, New Zealand

MICHELLE WU, Augmented Human Lab, ABI, The University of Auckland, New Zealand

SURANGA NANAYAKKARA, Augmented Human Lab, Department of Information Systems & Analytics, School of Computing, National University of Singapore, Singapore



Fig. 1. Troi is a mobile App that can recognise emotion with physiological signal captured from a wrist-worn wearable sensor. This was achieved by adapting a state-of-the-art machine learning model that runs within the mobile app. We deployed it in a real-world setting to investigate user perspectives to automatic emotion recognition application.

Authors' addresses: **Vipula Dissanayake**, vipula@ahlab.org, Augmented Human Lab, Auckland Bioengineering Institute (ABI), The University of Auckland, Auckland, New Zealand; **Vanessa Tang**, vanessa@ahlab.org, Augmented Human Lab, ABI, The University of Auckland, Auckland, New Zealand; **Don Samitha Elvitigala**, samitha@ahlab.org, HCI BoDi Lab, School of Computer Science and Engineering, University of New South Wales, Sydney, Australia; **Elliott Wen**, elliot@ahlab.org, Augmented Human Lab, ABI, The University of Auckland, Auckland, New Zealand; **Michelle Wu**, michelle@ahlab.org, Augmented Human Lab, ABI, The University of Auckland, Auckland, New Zealand; **Suranga Nanayakkara**, suranga@ahlab.org, Augmented Human Lab, Department of Information Systems & Analytics, School of Computing, National University of Singapore, Singapore, Singapore.

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Emotional Self-Awareness (ESA) plays a vital role in physical and mental well-being. Recent advancements in artificial intelligence technologies have shown promising emotion recognition results, opening new opportunities to build systems to support ESA. However, little research has been done to understand users' perspectives on artificial-intelligence-based emotion recognition systems. We introduce Troi, an automatic emotion recognition mobile app using wearable signals. With Troi, we ran a multi-day user study with 12 users to understand user preference parameters, such as perceived accuracy, confidence, preferred emotion representations, effect of self-awareness of emotions, and real-time use cases. Further, we extend our study to evaluate the machine learning model in-the-wild to understand behaviours in-the-wild. We found that users perceived accuracy of the emotion recognition model is higher than the actual model prediction accuracy; there was no strong preference for one specific emotion representation, and users' self-awareness of emotions improved over time.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; *Empirical studies in ubiquitous and mobile computing*; • **Computing methodologies** → Machine learning.

Additional Key Words and Phrases: Physiological Signals, In-the-wild Study, Automatic Emotion Recognition, Mobile Application

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1 INTRODUCTION

Emotional Self-Awareness (ESA) is the first component of emotional intelligence. It plays a vital role in many aspects of life, including interpersonal relationships [50], mental and physical health [45, 58], work performance [9] and academic well-being [44]. Emotion journaling is the most common tool used to track and improve ESA [29]. There exists a wide range of emotion journaling applications. However, most of them rely on users to self-report their emotions [51]. However, self-reporting could be a burden to end-users with their daily activities [3].

Recently, many researchers have been experimenting with automated emotion journaling to overcome the drawbacks of self-reporting. Specifically, they attempt to infer human emotions by applying machine learning technologies to different sensory data readings, such as facial images [40], vocal recordings [21], and physiological responses [7, 20, 21, 31, 77]. Using physiological responses for automatic emotion recognition has drawn considerable attention from researchers. Unlike other emotional indicators (e.g., facial [40] and vocal [21]), physiological responses are less affected by human subjective consciousness and behaviour [53]. Connecting smart wearable devices to mobile phones, we can now capture physiological signals continuously and unobtrusively, enabling automatic emotion recognition [57]. Further, physiological signals have fewer privacy concerns than audio and video signals.

Despite the progress in research around different algorithms to recognise emotions [7, 20, 31, 60] and some applications [18, 34, 76] demonstrating positive effects of emotion journaling from diverse research communities, limited research [2, 46, 72] explores the human perspective of such technology. Research indicates that most of the existing apps focus on tracking and reflecting emotions, and a limited number of apps provide features to take action based on emotions [52]. Further, existing research suggests that some end-user concerns are confidence, privacy, and system accuracy [2]. Yin et al. [73] report that people's confidence in a machine learning system depends on the accuracy they perceive from the system. According to Papenmeier et al. [47], perceived accuracy could significantly differ from the calculated accuracy. Thus when evaluating an automatic

emotion recognition method in real-life, performance matrices such as confidence and perceived accuracy should be considered.

Further, when a machine learning model is deployed, its' performance may not be similar to how it performs with a hold-out dataset. This effect is called the "concept and data drift" [17, 24, 73]. Hence, identifying the effect of concept drift for user perspectives in an in-the-wild deployment will benefit the broader adaptability of automatic emotion recognition systems.

In this paper we aim to understand end-users perspectives to an automatic emotion recognition mobile application in their natural setting. In particular, we aim to answer the following research questions.

- (1) What is the agreement between users' perceived emotions prediction accuracy and our model prediction accuracy?
- (2) How do emotion recognition model trained and evaluated with hold-out dataset perform in the wild?
- (3) Is there a user preference for emotion representation (categorical vs dimensional)?
- (4) How does the automated emotion recognition system influence users' emotional self-awareness?
- (5) What are user perspectives for automated emotion recognition systems?

In order to answer the above research questions, we developed Trois, a mobile app that can predict emotions based on the physiological signals captured from wearable devices. As this paper's aim was not to develop an emotion recognition model, we selected state-of-the-art models from the literature for our mobile application.

We conducted a multi-day user study with 12 participants, whereby we invited participants to use Trois throughout their day while they went about their everyday activities. We conducted semi-structured interviews with our participants after the deployment to understand how a system like Trois can be improved from the end-users perspective.

The study has revealed that users have higher perceived accuracy than the actual model prediction accuracy, and Trois users have significantly improved their self-awareness of emotions.

In summary, we make the following contributions.

- Implementation details¹ of the mobile app with a built-in machine learning model for emotion recognition.
- Empirical findings and insights from the user study to answer the research questions outlined above.

2 RELATED WORK

2.1 Emotion Tracking in HCI

Although there are many commercial emotion tracking apps (e.g. Daylio², eMoods³), most apps are not scientifically studied [1, 5]. Here we take a close look at the apps that have been evaluated. Emotion Map [34] and MobiMood [18] are two applications where users can log and share their emotions with friends. In addition to the common finding of improving emotional awareness in the users, Church et al. [18] report an increment in communication between users. However, Huang et al. [34] identified that the app should not only rely on people "sharing" emotions. It should focus on "logging" emotions. On the other hand, some applications such as CopeSmart [39] also have the additional feature of promoting positive coping strategies. Some applications were developed to assist in improving mental health. Indeed, MoodPrism [5] provides evidence for a decrease in depression and anxiety and an increase in mental well-being from studying 234 users.

¹<https://github.com/vipulaSD/Trois>

²<https://daylio.net/>

³<https://emoodtracker.com/>

Longitudinal studies have also been conducted; Alslaity et al. [1] examine self-report entries on the app called “Moodie” for two years and evaluated how factors such as location and season could affect moods. Despite the positive results of these applications, as mentioned in earlier studies, one major limitation is the dependency on self-reporting. Self-reporting could be a significant burden to end-users, especially when they engage in daily activities [3]. Despite self-reporting of emotions, Chen et al. [15] attempt to improve emotional awareness through a reflection questionnaire and guided prompt via a web-based application. However, their study results suggest no significant improvement in emotional awareness. Further, their system requires end-users attention to answer the questionnaire, which could be a burden. In order to overcome the limitation of the requirement for the end-users attention, machine learning algorithms could be used to recognise emotions automatically.

TapSense [28] is one of the automatic emotion recognition examples. It introduces a personalised algorithm to identify emotions with the typing characteristics of users. Despite a high accuracy in recognising emotions, their algorithm can only make predictions when the users are using the mobile keyboard. The dependency on keystrokes restricts emotion recognition when users are not engaged with the mobile keyboard. McDuff et al. [43] explore the machine learning algorithms in sensed data such as facial actions and expressions, posture features and electrodermal activity (EDA). However, this is a system constrained to a desktop computer set-up. Similarly, Kaur et al. [38] have also studied a desktop based automatic facial emotion recognition tool in a workplace context. Their results suggest a mismatch between internal emotion and external-facing emotion. With the online space dominating human interaction, some studies have explored and evaluated different automatic emotion recognition systems for online interactions. Nguyen et al. [46] built Moody Man, a text-based emotion recognition system run on Slack⁴, and have shown that the tool could help users improve shared understanding and creativity in a virtual team. Rojas et al. [55] have deployed an audio and visual-based emotion recognition system with video conferencing and reports valence reflection, which has enhanced users’ empathy. A detailed review of automatic emotion recognition methods can be found in Section 2.4.

2.2 User Perspectives of Automatic Emotion Recognition Apps.

Although a plethora of emotion recognition research and applications are based on sharing emotion, only a limited number of work has explored the user perspective on these tools [2, 4, 46, 66, 72]. In a recent work by Andalibi and Buss [2], they investigate people’s perspective of emotion recognition on social media by discussing over interviews an “*imaginary*” scenario where social media platforms use emotion recognition techniques on participants’ social media posts. Researchers discover that people have concerns over privacy, consent agency and potential harm. Wang et al. [72] have explored students’ perspectives on an affect sensor to support education. Although the main focus of the study is on designing the sensors, in the interviews, participants have revealed their thoughts on emotion recognition. These participants indicate that the primary technological concerns are the accuracy of affective state recognition and data privacy. In a study of a text-based automatic emotion recognition system for the workplace, Nguyen et al. [46] have found out that user’s trust plays a significant role in using a machine learning based affect recognition system. Furthermore, the authors highlight that further research is required in emotion representation. After reviewing commercially available mood-related apps, Ptakauskaite et al. [52] highlight that existing apps provide features to reflect the mood but provide no features to take further actions.

Some research work also exists to understand user perspectives on emotion recognition applications outside the HCI community [4, 12, 22]. Those works focused on reviewing apps that

⁴<https://slack.com/>

target mental health and having an impression in favour of self-reporting mainly because of the lack of trust in machine learning approaches. However, recent breakthroughs in machine learning techniques, affective computing and omnipotent of multiple sensors suggest the technology could have changed dramatically by now. Therefore, further research is required to understand user perspectives of automatic emotion recognition applications powered by one or a combination of current technology.

2.3 Perceived Accuracy

In the previous subsections, it was highlighted that emotion prediction accuracy is one of the main concerns from the user's perspective. Here, we define "accuracy": when a machine learning model is objectively evaluated, accuracy is the percentage of the correct prediction made by the model over the total number of predictions. However, when a machine learning model is deployed, the model performance is assessed by the users as they experience the system. This user-assessed accuracy is defined as "*Perceived accuracy*". A study by Papenmeier et al. [47] revealed that the perceived accuracy could be significantly different (lower or higher) from the calculated accuracy. Yin et al. [73] report that users' perceived accuracy plays a vital role in the trust users have in the machine learning system. Thus we suggest that users perceived accuracy is an important metric to evaluate a machine learning system from the users' perspective.

2.4 Emotion Recognition Methods

Different signal modalities have been explored to recognise the human affective state. These include audio [14, 21], visual [7, 41] and physiological signals (e.g., heart rate [20, 65], electroencephalogram (EEG) [31, 67], electrocardiogram (ECG) [33, 36], electromyogram (EMG) [16, 49], electrodermal activity (EDA) [20, 31]), touch screen interaction patterns [28, 71] and body interactions [26, 27, 37]. The advantage of these modalities is the continuity of signal capturing and emotion capturing. However, deploying some of those technologies in real life has practical difficulties. For example, capturing EEG, EMG and ECG signals requires a complex setup [75]; using a camera or mic to continuously capture signals has major privacy concerns [23, 75]. Camera-based signal capturing is also severely affected by light conditions [23]). In the meantime, the recent advancement of smart wearables opens up an opportunity to use physiological signals for emotion recognition models. Although wearable signals can omit some practical difficulties discussed previously, they have inherent drawbacks. Frequent signal losses, noises and differences in the sensors make it challenging to use wearable signals for continuous emotion recognition [13]. The current state-of-the-art research, such as SigRep [20], has shown promising results in overcoming those difficulties with wearable signals. However, the effect of the real-world deployment of these systems, such as SigRep [20], is yet to be understood.

2.5 Emotion Representations

Emotions are complex phenomena experienced by humans. Studying human emotion is now a multidisciplinary field encompassing psychology, affective computing and computer science. In the research domain, emotions can be represented by different representations [25, 30, 35, 56]. These representations can be divided into two main branches, categorical and dimensional representation. In the categorical representation model, emotions are presented as a single label, such as happy, sad and angry. Although there are multiple categorical emotion representations [30, 35], they are derived from basic emotions [25]. In the dimensional representation model, emotions are represented as a point in a multi-dimensional space [56]. The dimensions are valence (positive vs negative) and arousal (high vs low). The Self-Assessment Manikin (SAM) scale [8] is a widely used self-report tool in dimensional emotion representation [6, 31]. However, the SAM scale was

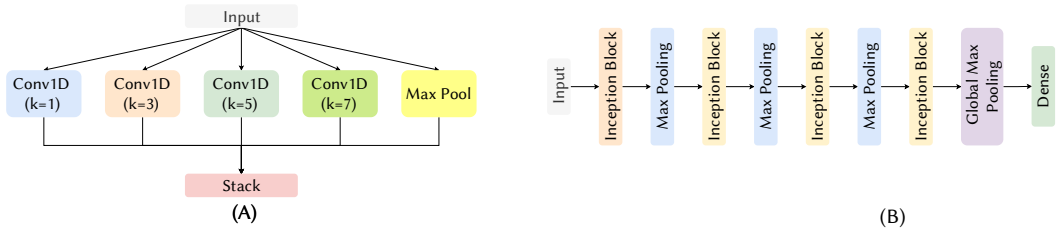


Fig. 2. SigRep [20] Model Architecture. (A) Inception block: A block of neural network built with 1D convolutions. Number of each convolution is represented in k . (B) Final model architecture built with Inception Blocks.

designed initially as a paper-based self-reporting tool [6]. Alternatively, researchers have proposed that AffectButton [10], Affective Slider [6], and EmojiGrid [70] be used digitally. The body of emotion recognition research has used either categorical [41, 65, 74] or dimensional [21, 31, 67, 75] representation or both [20, 33]. There is no explicit agreement as to which representation model to use. Therefore researchers and practitioners choose one which they think to be most relevant to their application [11]. However, when building a system targeting the general population, the end-users preference and understanding of representation play a significant role.

In summary, current research work and commercial applications of emotion self-tracking heavily depend on self-reporting emotions. Increased usage of digital devices and the advancement of machine learning techniques have enabled an opportunity to replace those active emotional self-reporting mechanisms with passive automatic emotion tracking systems. However, understanding user perspectives in this automatic emotion logging system is limited. This paper aims to contribute to this knowledge gap by investigating user perspectives such as perceived accuracy, the confidence of model predicted accuracy and preference for emotion representation.

3 TROI

Troi is a mobile platform that can recognise emotions or emotional states (categorical or dimensional representation) by interpreting physiological signals from a generic wrist-worn smart device. Note that the primary focus of Troi is to collect user responses to automatically-detected emotions and understand if the emotion representation may impact user perceptions. Troi can predict emotions in both categorical emotion representation [25] and dimensional emotion representation [56] models.

3.1 Emotion Recognition Model

We adapted a state-of-the-art method called SigRep [20] from recent work. SigRep [20] is a self-supervised-learning based method for emotion recognition using wearable signals. Authors of the SigRep [20] have shown that their method had achieved the state-of-the-art emotion recognition performance for multiple datasets and higher robustness against signal losses. Further, they have introduced a lightweight model architecture suitable for resource-constraint environments such as smartphones. We want Troi to run offline on a smartphone, and Troi depends on signals captured from a wearable device where we anticipate signal losses. Thus, using the SigRep [20] model and method would help us run Troi on smartphones and also mitigate the complications caused by signal losses.

We implemented the model described in the SigRep [20] using Python and Tensorflow v2.8⁵. The SigRep [20] model architecture has followed similar design principles to the inception architecture [69] for computer vision. Figure 2 (B) visualises the lightweight model architecture proposed in SigRep [20], constructed with the building block visualised in Figure 2 (A).

The training process introduced in the SigRep [20] has two major stages: (i) individual signal representation training and (ii) training for emotion recognition. In the representation training stage, a large set of unannotated signals from different datasets are used to train a neural network model to represent an individual signal. In SigRep [20], neural network representations are trained for the accelerometer (ACC), blood volume pulse (BVP), electrodermal activity (EDA) and skin temperature (TEMP) signals. In the emotion training stage, the trained signal representations were used to construct multimodal emotion recognition models, which were further trained to recognise emotion targets in multiple datasets in a supervised manner.

Following SigRep [20], we collected all the datasets used for representation and emotion training AffectiveROAD[32], CASE [64], CLAS [42], K-EmoCon [48], PPG-DaLiA [54] and WESAD [59]. Following the procedure in the SigRep [20], we preprocessed the datasets. We resampled each signal to specified sampling frequencies as follows: ACC (32Hz), BVP (64Hz), EDA (4Hz) and TEMP (4Hz). Then the time-series data is chunked into four-second windows with one-second overlap. We initially trained individual signal representations for ACC, BVP, EDA, and TEMP using the preprocessed datasets, with the contrastive learning based method proposed in SigRep [20]. Later, we used pre-trained signal representations to construct emotion recognition models. Then following SigRep [20], we trained emotion recognition models in a fully supervised manner using CASE [64] and K-EmoCon [48] datasets.

Different emotion recognition datasets have represented emotion differently; some have used dimensional representation (arousal-valence), and some have used categorical representation. Sometimes emotional dimensions have represented in two stages (e.g., high-arousal, low-arousal) and sometimes in three stages (e.g., high-arousal, neutral, low-arousal). To address those kinds of scenarios, the authors of SigRep [20] have trained multiple models, just changing the number of units in the final layer of the emotion recognition model. For Trois, we selected the three-class arousal, valence prediction models and the categorical emotion prediction model from SigRep [20].

Until this point, we have followed the procedure proposed in SigRep [20] and selected three emotion recognition models to predict arousal, valence and categorical emotions. Those three models are built with Tensorflow. Tensorflow provides a lighter version called Tensorflow lite⁶, optimised for resource constraint devices such as smartphones and microcontrollers. Since we plan to deploy Trois on smartphones, we decided to convert our models to Tensorflow lite. This conversion allows us to run SigRep [20] models on a smartphone for a long duration. We used the Python API of the TensorFlow Lite converter⁷ for the model conversion process.

3.2 Mobile App Development

We developed the Trois app for the Android operating system. The app can communicate with an Empatica E4 device⁸ using the E4 Android SDK⁹ over Bluetooth. Figure 3 visualises the main components and the workflow of the app.

The app is designed to run mainly in the background. Once an Empatica E4 is connected, Trois will start a background service to listen to the sensor data stream. In addition, the service loads three

⁵https://www.tensorflow.org/api_docs/python/tf

⁶<https://www.tensorflow.org/lite>

⁷<https://www.tensorflow.org/lite/convert>

⁸<https://www.empatica.com/research/e4/>

⁹<https://developer.empatica.com/android-sdk-tutorial-200-legacy.html>

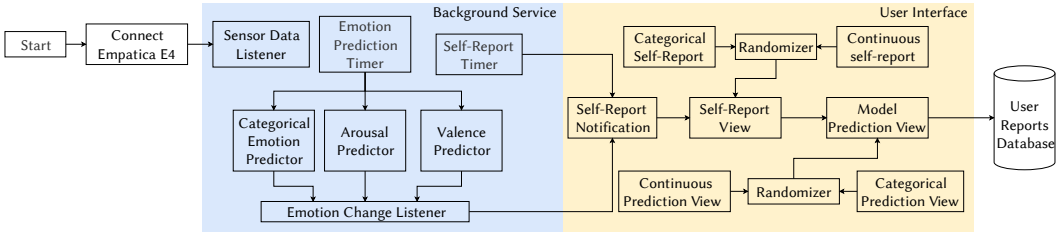


Fig. 3. Workflow of the mobile app

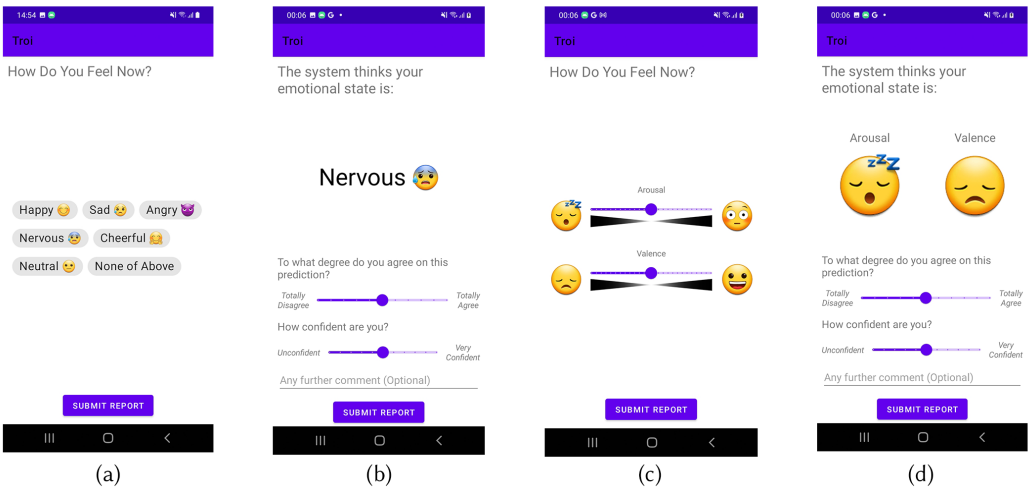


Fig. 4. Mobile User Interfaces for self-reporting emotions and providing feedback on the Troi predicted emotions. (a) Self-report emotions in categorical representation. (b) Predicted emotion in categorical representation. (c) Self-report emotions in dimensional model with Affective Slider. (d) Predicted emotion in dimensional representation.

emotion recognition models upon its initialisation. A periodic task, named the emotion prediction timer, is executed every minute. It predicts emotion in real-time using the sensor data stream during its execution. Then the predicted emotions are fed into an emotion change listener component, which monitors sudden emotional changes. If a sudden change is detected, it pops a notification to request a self-report from users. The Troi background service contains another periodic task called a self-report timer, executed every 20 minutes. Its task is to pop a self-report notification only if the emotion change listener has not popped the notification in the past 20 minutes. When a user clicks on the self-report notification, the app directs the user to the User Interface (UI) component.

3.3 Application Workflow

Troi has two main UI components. (i) UI for self-reporting emotions, (ii) UI for rating system predictions. Each UI component has two variants for different emotion representations. We use

two UI variants to examine the effect of emotion representations. Figure 4 shows both variants of two major UI components.

The UI component for self-reporting emotions prompts users to input their perceived emotions. The app shows six emotions (happy, sad, angry, nervous, cheerful, and neutral) in selectable chips¹⁰ in the categorical emotion variant. Each emotion chip shows an emoji that represents the emotion label. The app also allows users to provide a custom emotion label by selecting the "None of above" option. In the dimensional emotion variant, the app asks users to report their emotions in an arousal value and a valence value between 0 and 1. We use the Affective Slider method [6] to implement the arousal-valence self-report UI.

After the user self-reports the perceived emotion, the app displays the model's predicted emotion. The system prediction UI component also has two variants based on the emotion representation. Even though the models predict emotion in both representations, we randomly select one representation to understand the effect of the representation. The app visualises the categorical emotions as the emotion label followed by an emoji representing the emotion. Troi visualises emoji representing arousal and valence levels separately for arousal-valence emotion representation. In each scenario, we ask the user to rate their agreeability and confidence in the system on a scale between 0 (completely disagree) to 10 (completely agree). Once a user has rated the system prediction, we store user rating, self-reports and model prediction in a cloud database¹¹ for further analysis.

4 EVALUATION

4.1 Aims

In this study our goal was to evaluate the following research questions.

- RQ1: What is the agreement between users' perceived emotions prediction accuracy and our model prediction accuracy?
- RQ2: How do emotion recognition model trained and evaluated with hold-out dataset perform in the wild?
- RQ3: Is there a user preference for emotion representation (categorical vs dimensional)?
- RQ4: How does the automated emotion recognition system influence users' emotional self-awareness?
- RQ5: What are user perspectives for automated emotion recognition systems?

4.2 Participants & Procedure

We recruited 12 participants (3 Females and 9 Males, aged between 18 and 45 ($M=30.14$, $SD=4.15$)) to conduct a four-day in the wild user study. Figure 5 summarises the overall procedure of our user study. First, we conducted a 15-minute one-to-one introductory session with each participant. We explained the study's purpose and demonstrated how Troi and Empatica E4 work. Participants were then asked to complete a questionnaire with demographic questions and the Schutte Self-Report Emotional Intelligence Test (SSEIT) [62]. The SSEIT comprises 33 questions to measure general emotional intelligence. Each question needs to be answered on a five-point Likert scale (1-> strongly

¹⁰<https://material.io/components/chips>

¹¹<https://firebase.google.com/docs/firestore>

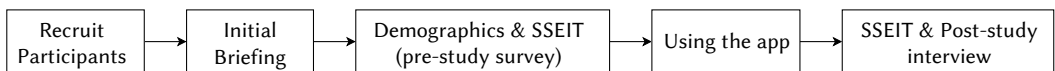


Fig. 5. Workflow of the study.

disagree, 5-> strongly agree). Our questionnaire also contains open-ended questions about their previous experiences using mobile applications to maintain their emotional well-being.

After the introductory session, the participants were requested to use the application for four days. During these four days, Troi displayed a prompt periodically (i.e., every 20 minutes) or when Troi detected a notable change in an emotional state. Each prompt contained two tasks: 1) participants were asked to self-report emotions at that moment, 2) the app then displayed a predicted emotion, and the participants were asked to report how agreeable they were and their confidence in the predicted emotion by Troi (on a scale of 0 to 10). Troi randomly selected one of the two emotion representations in those two tasks: categorical or dimensional. We used the following categorical emotions: happy, sad, angry, nervous, cheerful and neutral. The dimensional model represents emotion in arousal and valence domains (see Figure 4).

While participants were encouraged to answer the prompts as often as possible, we did not enforce any usage restrictions. The study was conducted in the wild while users engaged in their day-to-day activities. At the end of the four days, we asked the participants to fill in SSEIT [62] again, and we conducted a semi-structured interview with each participant. In the interview, we discussed the overall experience with Troi and the potential use cases of Troi in their day-to-day life. We followed up with any interesting points that emerged in their answers. The overall study workflow is shown in Figure 5.

4.3 Data Analysis

In order to answer our research questions, the following analyses were conducted:

User agreement and confidence on emotion recognition system (RQ1): The user agreement and confidence were reported on a Likert scale, from 0 (totally disagree/no confidence at all) to 10 (totally agree/very confident) for each prediction. We break down our analysis into the per-user level to observe any personal effect. This breakdown would also allow us to gain insights into an individual's perspective on the system. Furthermore, we examine the agreement and confidence scores for the two cases: when the user's self-reports match with model prediction and when they do not. Before running the statistical analysis, we test for the normality of the data with the Shapiro-Wilk test. Based on the normality test results, we used Wilcoxon signed-rank test to identify statistical significance in our results.

Emotion recognition performance in the wild (RQ2): We evaluate the performance of the emotion recognition model using the user self-report emotion label in task 1. The performance metric used is prediction accuracy, the percentage of model prediction that matches with self-reported emotion label. We compare our results (data collected in-the-wild) with relevant accuracy values reported in the SigRep [20], which had been evaluated with a hold-out dataset.

Choice of a emotion representation model (RQ3): We test for normality with the Shapiro-Wilk test on the self-reported agreement and confidence data for the two scenarios (i.e. when Troi present the predicted emotion using categorical and dimensional representations). We conducted the Mann-Whitney U test to compare two scenarios based on the Shapiro-Wilk test. In addition, we extend our analysis to the per-user level to examine any individual effect. Lastly, we qualitatively summarise post-study interview data on the user preference for the emotion representation models.

Troi's effect on users emotional self-awareness (RQ4): To see any effect of the user study on the participants' emotional intelligence, we analyse pre-study and post-study SSIET [62] scores for each participant. Thirty-three questions in the SSEIT [62] comprise four subscales [61]: perception of emotions, managing own emotions, managing others' emotions, and utilisation of emotions. For each question, users have to provide a score on a scale of 1 (totally disagree) to 5 (totally agree). Out of the four subscales, we choose to analyse the "perception of emotions" subscale as it is most relevant to our research question of emotional self-awareness. The subscale consists of ten

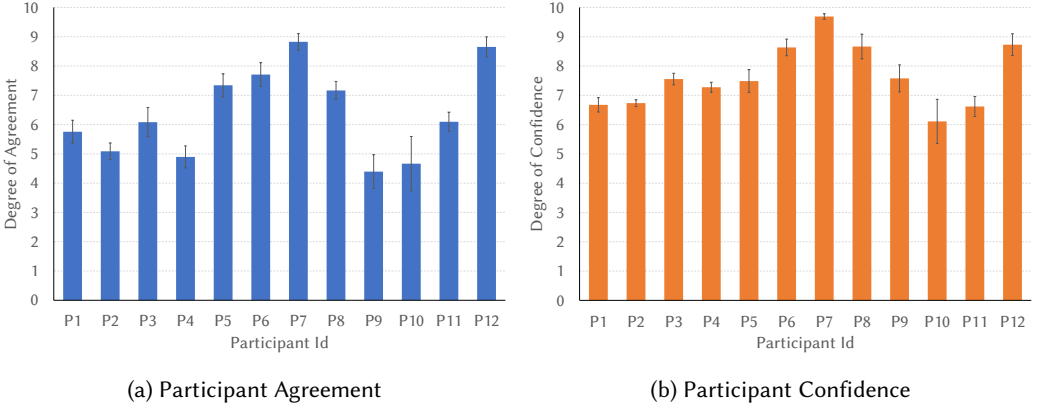


Fig. 6. Participant agreement and confidence on system prediction.

questions. We average user-provided Likert values to calculate the subscale score. Similar to the previous analysis, we first test for the normality of data with the Shapiro-Wilk test and select the T-test to test for statistical significance.

Future directions for automated emotion recognition systems from the users perspective (RQ5):

We audio-record all the post-study semi-structured interviews. These audio records are transcribed and thematically analysed [19, 68]. We explore the possible directions for automated emotion recognition systems from an end user's perspective from the main points that surfaced from the interviews.

5 FINDINGS & DISCUSSION

We present our findings inlining with each research question as follows.

5.1 Users perceived accuracy was higher than the actual accuracy of model prediction predictions

Overall, participants reported an agreement score of $6.56 \pm 0.15(SE)$ and a confidence score of $7.80 \pm 0.10(SE)$, although our model achieved an accuracy of only 52.91% compared to users' self-report. The overall accuracy was calculated as the percentage of exact match of model prediction and users' self-report. This result indicates that, generally, participants have perceived above-average agreement and confidence with the model prediction.

Before further analysing the agreement and confidence scores, a Shapiro-Wilk test was performed to identify the normality of the data. Test results indicated that both agreement ($W_{agreement} = 0.916, p < 0.001$) and confidence ($W_{confidence} = 0.887, p < 0.001$) scores were not normally distributed. Therefore, we used a non-parametric Wilcoxon signed-rank test to compare the agreement and confidence with the neutral (5.0). Pairwise Wilcoxon signed-rank test confirmed that users have significantly higher agreement ($Z_{agreement} = 9.574, p < 0.001$) and significantly higher confidence ($Z_{confidence} = 15.665, p < 0.001$) with Trois compared to the neutral (5.0).

There are few possible explanations for this observation of higher perceived accuracy than the actual accuracy. Zhang et al. [76] suggest human emotion could be a combination of basic emotions, identified as a compound emotion at a given time. In the case of Trois, our emotion recognition model predicts one basic emotion in a given time, and the users self-report one basic emotion at

any given time. The accuracy of our model is calculated based on whether these two basic emotion values match. When we ask users to provide agreement and confidence on the model prediction, they might contemplate the prediction. If the prediction were a part of the compound emotion users had at the moment, they would agree with the model to some degree. This situation could lead to a higher perceived accuracy than the calculated model accuracy.

Yin et al. [73] have found that people are likely to increase trust in a machine learning system when the model's observed accuracy is higher than their own. Also, the authors suggest simple interventions such as letting people make predictions themselves before seeing the model's prediction would encourage users to evaluate their prediction against the model. The term "trust", used by Yin et al. [73], would be reflected by the users' degree of confidence in our work. In Troi, users indeed see the model prediction after their self-report. As we discussed previously with the theory of compound emotions, a person can experience multiple basic emotions simultaneously. When our model predicts one of the multiple emotions, regardless it matches with the self-reported one, users perceive the model as accurate to some degree. When the predicted emotion was one of the compound emotions but not the self-reported emotion, users might reassess their judgement and view the model prediction as more accurate than their own. Thus, according to Yin et al. [73], users would have more trust in the system. This finding explains why Troi got higher than the neutral confidence from the users. Furthermore, during the interview, P3 has mentioned "*I think at least like 80% of the time they [predictions] were very close to what I marked. That [Troi] was accurate more than I expected.*". They perceived a higher accuracy than expected could also have led users to provide high confidence scores.

We further analysed the level of agreement and confidence for each participant. Figure 6a and Figure 6b illustrate per participant agreement and confidence respectively. The Wilcoxon signed rank test indicates that P5 ($Z_{\text{agreement}} = 4.343, p < 0.001$), P6 ($Z_{\text{agreement}} = 4.833, p < 0.001$), P7 ($Z_{\text{agreement}} = 6.142, p < 0.001$), P8 ($Z_{\text{agreement}} = 3.624, p < 0.001$), P11 ($Z_{\text{agreement}} = 2.771, p = 0.006$), P12 ($Z_{\text{agreement}} = 4.499, p < 0.001$) have a significantly higher ($p < 0.05$) agreement than the neutral (5.0). Further, every participant P1 ($Z_{\text{confidence}} = 4.392, p < 0.001$), P2 ($Z_{\text{confidence}} = 5.796, p < 0.001$), P3 ($Z_{\text{confidence}} = 5.147, p < 0.001$), P4 ($Z_{\text{confidence}} = 4.7, p < 0.001$), P5 ($Z_{\text{confidence}} = 4.451, p < 0.001$), P6 ($Z_{\text{confidence}} = 6.33, p < 0.001$), P7 ($Z_{\text{confidence}} = 6.687, p < 0.001$), P8 ($Z_{\text{confidence}} = 3.446, p < 0.001$), P9 ($Z_{\text{confidence}} = 3.678, p < 0.001$), P11 ($Z_{\text{confidence}} = 3.13, p = 0.002$), P12 ($Z_{\text{confidence}} = 4.444, p < 0.001$) have demonstrated significantly higher ($p < 0.05$) confidence than the neutral (5.0) confidence.

We further analysed the agreement and confidence values when Troi predictions match user self-reports and when they are not matching. We plotted the results in Figure 8. As the agreement and confidence data is not normally distributed, we used the Mann-Whitney U test to analyse two samples. We found a significant decrease of agreement ($U_{\text{agreement}} = 7.493, p < 0.001$) and confidence ($U_{\text{confidence}} = 4.648, p < 0.001$) when model prediction was clearly different from the self-reports.

According to Nguyen et al. [46] perceived accuracy plays a major role in end-users trust and confidence in a machine learning system. Papenmeier et al. [47] has found out that not all incorrect predictions affect equally to reduce perceived accuracy (i.e. obvious errors have a higher impact). Therefore improving the perceived accuracy can gain end-user trust in a machine learning system. Although Troi has shown overall higher perceived accuracy than actual, the breakdown suggests there is still room to improve the models. This observation calls for further research efforts on better emotion prediction models.

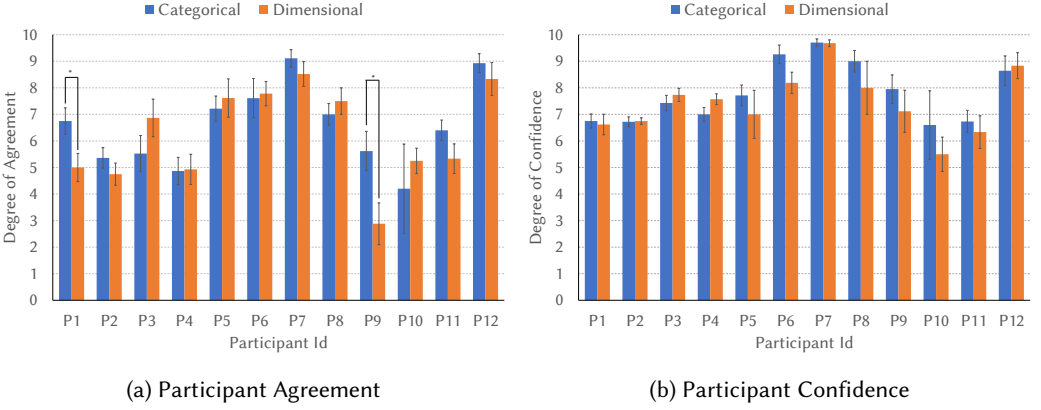


Fig. 7. Effect of Emotion Representation on Participant Agreement and Confidence

5.2 A model trained on a dataset performs on par when it is deployed in the wild

Overall, our models have predicted participants' self-reported emotions with an accuracy of 52.91%. If we broke down the accuracy based on the underlying categorical and dimensional model, we observed that participants' self-reported emotion category matches the model predicted emotion category 56.19%, and Troi has predicted valence and arousal levels with an accuracy of 64.36% and 65.35% respectively. In SigRep [20], authors have reported accuracies for multiple emotion recognition tasks based on the emotion labels provided in different datasets. We compared the most comparable results reported in SigRep [20]. These are 51.21% for categorical emotion recognition, arousal (65.07%) and valence (64.83%) prediction in three levels.

It is possible for a machine learning model trained on a hold-out dataset to demonstrate inferior performance when deployed in the wild [17, 24, 73]. It happens due to the mismatch between the trained dataset and in-the-wild data distribution. However, in our case, Troi has performed on par with the evaluations conducted with the fixed dataset. It implies the models and method proposed in SigRep [20] have a superior generalising capability beyond the distribution of the datasets it was trained on. However, in practice, if a machine learning based model is deployed in the wild, it is recommended to monitor the system performance over time and take necessary actions to improve model performance, such as re-training the model periodically with recent data or through on-device personalisation based on the user's data over time [24].

5.3 Emotion representation has limited impact on user agreement and confidence

Overall, when predicted emotion is visualised in categorical representation, participants have reported an agreement value of $6.74 \pm 0.19(SE)$ and a confidence value of $7.87 \pm 0.13(SE)$. In the scenario where dimensional representation is used, participants have reported an agreement of $6.35 \pm 0.22(SE)$ and a confidence value of $7.69 \pm 0.16(SE)$. Since the agreement and confidence values are not distributed normally, we conducted Mann-Whitney U tests to identify significance of agreement ($U_{agreement} = 1.275, p = 0.202$) and confidence ($U_{confidence} = 0.507, p = 0.612$) values for each representation. We do not observe any significant difference between the two representations. We further analyse data on a per-participant basis to examine individual preferences for specified representations. Figure 7a and Figure 7b illustrate per-participant agreement scores and confidence scores respectively. Most of the participants have not demonstrated a significant preference for

representations. However, we observe a significant difference for P1 ($U_{agreement} = 2.32, p = 0.021$) and P9 ($U_{agreement} = 2.45, p = 0.015$). In those cases, participants have demonstrated significantly higher ($p < 0.05$) agreement with dimensional representation. All users have not demonstrated a significant difference in confidence with respect to emotion representation.

Even though the user agreement and confidence report do not provide information on the impact of the emotion representation model, during the interview, users provided mixed preferences for the representations. P2 have mentioned that “*Actually, I had a tough time with those axes [arousal/valence] because I’m not used to.*”, P4 has mentioned that “*I like this scale [arousal/valence] a little bit more. Because I feel like I could probably match that [to compound emotion].*” and P11 have mentioned that “*I think, asking to report in both [categorical and dimensional] in predicting on both [categorical and dimensional] is probably a better idea*”.

Results suggest no solid evidence to support one specific emotion representation. We believe it is important to support multiple emotion representations and allow end-users to select their preferred representation in an application.

5.4 Using Troi has significantly improved the perception of emotions of participants

A comparison of pre-study and post-study SSEIT “perception of emotion” subscale is plotted in Figure 9. We conducted a Shapiro-Wilk test to test the normality of pre-study and post-study values. Test results indicate that pre-study SSEIT ($W_{pre-study} = 0.928, p = 0.362$) and post-study SSEIT ($W_{post-study} = 0.906, p = 0.190$) data is distributed normally. Therefore we used a pairwise t-test to compare data. Overall, we observed a significant improvement in SSEIT scores ($t_{11} = 2.212, p = 0.025$), meaning that our Troi users have improved their perception of emotion during the study period.

This significant improvement could be caused by users frequently contemplating their emotions during the study period. The contemplation could happen when users self-report, as well as when users provide agreement and confidence scores to a Troi prediction. Both stages require users to think about how they feel at the given time. Indeed, P6 reported that she “never thought about

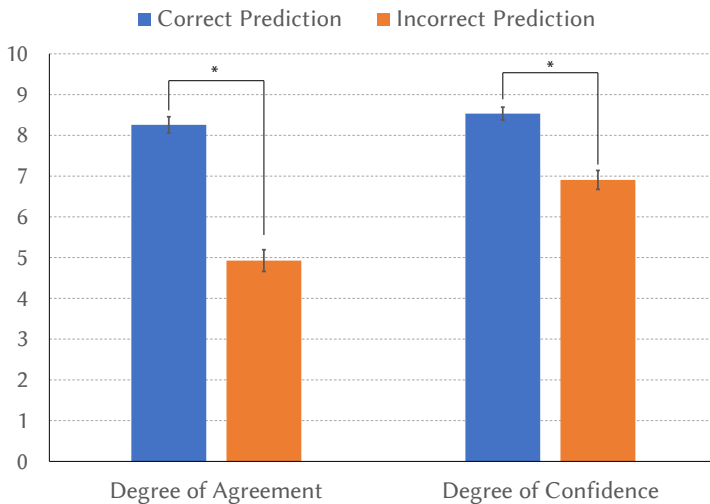


Fig. 8. User agreement and confidence scores when model prediction matches with self report.

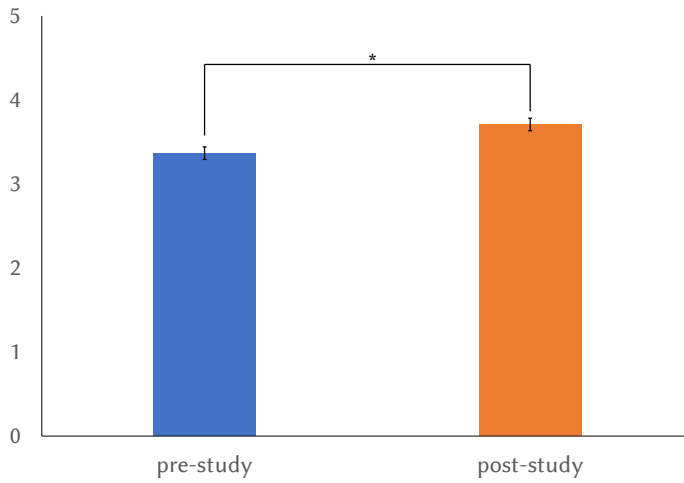


Fig. 9. SSIET: Emotion Perception of Emotion pre-study vs post-study. Higher scores implies better emotion perception.

what I was feeling before” and *P2* commented that previously he “think about my emotions not daily... maybe once a week”.

Results indicate that using Trois has helped users significantly improve their emotional perception, leading to emotional self-awareness. Literature suggests that the benefits of emotional self-awareness spread across multiple aspects of life [15, 63]. A majority of current work that empirically shows supporting for emotional awareness heavily depends on self-reporting of emotions [5, 18, 34, 39] which could be a hassle in real-life deployment [3]. We overcome the burden of self-reporting by automatically recognising emotions. The convenience accompanied by automatic recognition and favorable outcome in improving emotional self-awareness suggests automatic emotion recognition systems should be tools supporting health and well-being in real-life.

5.5 Towards Automatic Emotion Recognition System

In response to the question regarding user imagined use cases of this automatic emotional recognition system, most participants could brainstorm a number of ideas after they had hands-on experience with Trois. This section is organised around the main themes that emerged from the interviews.

Emotional Awareness: Many expressed the importance of understanding their own emotions as a way to “react better in [different] situations”. Therefore, they see the system as an “important tool to help people to identify their emotions”. As such, visualising how one’s emotion changes throughout the day and week is an important feature to many participants. *P3* and *P5* commented that having a tracking system for emotions, similar to other health and fitness apps, would be useful. Taking inspiration from those existing apps, *P2* had the idea of having a “emotional score” as a method to understand their own “emotional changing pattern”. Instead of the system reminding the user to log emotions at a specific time duration, *P1* felt reflecting on emotions after finishing an activity would be more useful, he imagined “an activity log with related emotions”. However, most mentioned the

need to have only one centralised application for fitness, health, stress and emotion reports, and the need to integrate the sensing method into an existing smartwatch.

User Agency: Most participants wanted to have control over the system. P5 emphasised the need to know when his physiological information was being tracked. Thus a notification was deemed necessary to inform the user when the smartwatch was disconnected. Similarly, P9 wanted a customisable system such that its activation and the information provided are dictated by the user – “*activate only for a period of time...sometimes I don’t need [system’s feedback], I just want to monitor.*”

Intervention Methods: For an automatic emotion recognition system, participants envisioned many intervention methods – “*recognition is only part of the problem... some kind of intervention is needed to make it a full package.*” Three participants envisioned the system actively assisting them in regulating negative emotions with personalised suggestions. P2 shared “*if I feel sad for a longer period, maybe this [system] can tell me: ‘OK, this is not healthy. You should go out and do something’.*” A few participants mentioned the meditation as an intervention method for low mood. P7 and P10 saw the need of interpersonal connections when the user is feeling sad – “*connect to contact history (social media or phone) and it will send a reminder to friends*” about the user’s current emotional state. Three participants envision an emotion recognition system to actively react to their current emotional state. Intervention methods include muting or not displaying less important notifications, playing music depending on the emotions, and changing the room light and ambiance accordingly to emotions when connected to an virtual assistance.

Other possibilities: In addition to personal usage, participants also brainstormed other use cases for such system. P10 saw the possibility of using such emotional recognition system in industries that involves very emotional settings such as the police force or military – the system would detect negative emotions such as stress, overwhelmed, or scared, that could negatively impact the task performance. Subsequently the system would inform the headquarters to send personnel replacement for the task. In addition, the system could also be applied to help autistic kids – “*the system can capture the situations around the kids... and show the face when people are getting angry, so kids see it and also they will be able to internalise it.*” These varying ideas show us that the participants see value in such a system, not only for personal benefits, but also for the wider society.

6 LIMITATIONS & FUTURE DIRECTIONS

Dependency on self-reports: Throughout the multi-day study, our participants had to frequently self-report their emotions and agreement with the model predicted emotion when prompted. Participants may miss the self-reporting requests as they get on with their daily schedules. We started the study with a thorough introduction to reduce these possibilities and addressed all the participants’ concerns. Further, we regularly checked in with our participants to encourage and remind them to use the app. A support phone number was also given to users so they could seek assistance anytime required.

Dependency on external devices: The application has been developed to run on the Android Operating System (OS). For the participants using smartphones with different OS, we provided an Android phone with the app installed. Also, we used Empatica E4 to capture signals. Some participants were not used to wearing smartwatches, and some had to take off their personal watch to participate in the study. Using unfamiliar devices may have affected participants’ overall experience with Troi. In future work, we plan to overcome these issues by developing applications to support multiple

OSs. We also aim to work with different commodity wearable devices such as Apple Watch¹² and Galaxy watch¹³.

Model accuracy: Although the prediction accuracy of AI models is not the main focus of this paper, we observe that our models were not perfectly accurate all the time. We observe a significant effect of the prediction accuracy on user agreement and confidence. We suggest experimenting with better models as they become available in future.

Participants from a narrow population: Due to COVID-19 constraints, all of our participants are from the same city, limiting our findings' generalisability across societies. However, to reduce the adverse effects caused by that, we selected participants born and raised in different countries and cities. For future works, we plan to conduct a study across a few cities or countries to establish whether there are any effects of living in different societies.

7 CONCLUSION

This paper presented Troi, an automatic emotion recognition system using wearable signals. We deployed Troi with 12 participants for a multi-day in-the-wild study to understand user perception of automatic emotion recognition applications. We did not introduce our own models to recognise emotions. Instead, we re-implemented models presented in the literature. The adopted models were only evaluated with a hold-out dataset. We make our study an opportunity to evaluate those models in-the-wild. Through the evaluation, we demonstrated that in-the-wild deployed models have performed on par compared to their performance evaluated with a hold-out dataset. Also, we found out that Troi has significantly helped the participants improve their self-awareness of emotions and generally perceive a higher accuracy than the accuracy of machine learning models. Further, participants have provided high confidence in the system, which could result from higher perceived accuracy.

Furthermore, we found that users prefer personalisation of the app experience and envision automatic emotion tracking applications to provide personalised guidance to regulate their emotions proactively. We envision that this study will inspire future research work and consumer applications of automatic emotion recognition.

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¹²<https://www.apple.com/apple-watch-series-7/>

¹³<https://www.samsung.com/us/watches/galaxy-watch4/>

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